3 Yr. Degree/4 Yr. Honours 2nd Semester Examination, 2024 (CCFUP)

Subject : Computer Science

Course: COMP2011 (MAJOR)

Time: 2 Hours

Full Marks: 40

The figures in the right hand margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* questions: 2×5=10

(a) What is type casting?

(b) What is string?

(c) How does "++a" differs from "a++"?

(d) What is union?

(e) What do you mean by operator precedence?

(f) What is pre processor directive?

(g) Write the purpose of fopen ( ).

(h) Differentiate between malloc ( ) and calloc ( ).

2. Answer *any two* questions: 5×2=10

(a) What is pointer? Explain call by value and call by reference techniques with the help of an example. 1+4

(b) Write a menu driven program to design a simple calculator. (use switch-case) 5

(c) Define data type. Discuss different data types present in C. 1+4

(d) Define structure. Write the difference between structure and union. What is keywords? 1+2+2

3. Answer *any two* questions: 10×2=20

(a) What is recursive? Explain its machanism. Write recursive functions for the following.

(i) Factorial of a suitable integer.

(ii) Reverse of a string. 1+3+3+3

(b) What is dynamic memory allocation? Explain it with a suitable example. Write a program using pointer that sorts a set of ten numbers. 2+3+5

(c) What is function? Write a program on C that adds two matrix. 2+8

(d) Write short notes on *(any one)* :

(i) File handling in "C".

(ii) Arrays of structures